

Project: AI Robot Arm

1. Document Management Department

1.1 History of Variations

Version	Date	Description of the change	Author
v0.1	24/02/2026	Original Version Idea Development and Planning	Milena Petrosyan
v0.2	05/03/2026	Technical department added Sensor Testing	Milena Petrosyan
v1.0	28/03/2026	Working and Completed Project	Milena Petrosyan

1.2 Document Status

- Ready for Display

1.3 Responsible

Role	Name	Duty
AI/ML Developer	Milena Petrosyan	Code and integration, programming
Hardware Engineer	Milena Petrosyan	Devices
Researcher	Milena Petrosyan	researcher

2. Brief description

AI Robot Arm: A robotic AI arm is a system that repeats the movements of the human hand in real time.

1. What Problem Are You Solving?

2. My project solves the problem of recognising human hand movements and reproducing them through a robotic system. In many cases, it is difficult for people to remotely control mechanical systems or demonstrate precise movements. The system we created uses a camera to track the movements of a person's hand and fingers, processes the data with the help of artificial intelligence, and transmits it to a robotic hand that replicates the same movements in real time.
3. Beyond simply mirroring movements, the system can also recognise specific gestures and display their names on screen. This can be useful in the development of robotics, education, gesture recognition, and intelligent control systems. The project also demonstrates how artificial intelligence, computer vision, and Arduino can be combined into a single unified system.

2. What new do you suggest?

I propose an intelligent system that combines a camera, artificial intelligence, and a robotic hand into a single project.

The distinctive feature of the project is that it can simultaneously both mirror movements and recognise specific gestures. For example, the system can understand particular hand positions and display their names on screen. All of this is implemented using the Python programming language, computer vision technologies, and an Arduino microcontroller.

Our project has potential future applications across a variety of fields:

- Physical heavy labour
- Medicine
- Underwater exploration

3. What result did you get?

As a result of the project, a working prototype has been created that is able to recognize the movements of the human hand in real time, repeat them through a robotic hand, and display the corresponding names on the screen for some gestures.

2 AI models were used in our system. The first model is designed for gesture recognition. It uses the camera to see the position of the hand, compare it to pre-learned data and determine the gesture shown. For this model, we used the KNN (K-Nearest Neighbors) algorithm, which compares the current hand position with pre-learned examples and selects the closest match. In the end, the system displays the name of the gesture on the screen.

The work of a model can be compared to an apprenticeship. If we show a student few examples or incorrect data, he or she will also learn wrong. For this reason, we have carried out preprocessing, cleaning up all the unqualified and incorrect images to get a more accurate result and a higher recognition quality.

The second model is based on the MediaPipe library. It places 21 dots on the hand and uses the movement of those dots to understand which finger is moving and the position of the hand. As a result, when the hand appears in front of the camera, these dots are visible on the screen and the system is able to track the movements of the hand and transmit them to the robotic hand.

Two datasets were used in the project:

- HandGesture Dataset, which contains about 23,000 different gesture images,
- FreiHAND Dataset, which contains about 20,000 hand drawings.

The labels matched the sign names, which allowed the model to learn the correct distinction between different gestures.

Our main goal was to make the system work in real-time so that the robotic arm responds to human movements as quickly as possible. It is also important to note that the robotic arm was made entirely from scratch.

To evaluate the effectiveness of the models, we used several criteria:

- Accuracy of gesture recognition
- the size of the average error

latency, that is, how quickly the system responds and transmits motion to the robotic hand.

During the experiments, we noticed that if the speed of movement is increased too much or the lighting is poor, the system sometimes recognises gestures incorrectly. This can be compared to reading for a fast-moving bus; if it passes too fast, it is possible to misread the number.

The end result is that the system is able to recognize about 50 different classes that correspond to different gestures. The robotic arm successfully repeats human movements in real time, and the system displays the names of recognized gestures.

3. Description of the project

✓ Idea

What is a Robot arm and how does it work?

The Process of Work

Our model accepts 21 points on the hand, and according to the position of those points, it understands which gesture we are making now. The resulting data is transmitted to the computer, and the computer sends it to the Arduino.

Discovery: The camera fixes a person's hand and transmits the image to the system.

Tracking and Analysis: The MediaPipe model places 21 points on the hand and uses them to understand the movements of the hand and fingers.

Gesture recognition: The KNN model compares the position of the hand to the gestures it has learned and determines the gesture that is displayed. The name of the gesture is then displayed on the screen.

Transmission of movements: The system transmits data to the Arduino, and the robotic arm repeats human movements in real time.

✓ Objectives

What did we want to get?

The main goal of our project was to create a robotic hand that would be able to recognize the movements of the human hand in real time and repeat them as accurately as possible. We also wanted the system to be able to recognize different gestures and display their names on the screen.

Accuracy: To create a system that will correctly understand hand and finger movements even with such gestures. To do this, we used AI models and 21-point analysis of the hand.

Optimization (speed): Our goal was for the system to work in real-time mode, i.e. the robotic arm would react as quickly as possible to human movements without much delay. For this reason, we also measured latency, how fast the system transmits motion to the robotic hand.

Applicability (Utility): We wanted to develop a system that could be used in robotics, training and gesture recognition systems. It is also important to note that the robotic arm was made entirely from scratch.

✓ Problem

Why have we chosen this topic and what problem are we solving?

Why this topic? I chose this topic because I am interested in artificial intelligence and robotics, and I carried out the whole project on my own.

My goal is to create a system that uses a camera to recognize the movements of a human hand and repeat them with a robotic hand, in real time.

The system solves the problem of motion recognition and simple human-robot interaction without complex control devices.

4. Device Integration

Arduino Uno – the main controller that receives data and controls the server engines

Camera – fixes hand movements in real time

MediaPipe (Python library) – a computer vision library that detects 21 hand points and tracks movements

OpenCV (Python library) – a library that helps you process camera images and work with real-time video

Scikit-learn – KNN (Python library/model) – a library used to recognize gestures by comparing hand positions and determining the appropriate gesture
MG996R servo motor – controls finger and hand movements
Stepper motor + driver – provides basic mechanical movements and smooth rotations

5. Technical Data + Technology

Device Data

Arduino Uno – small-sized microcontroller, low power consumption (~5V), works between 0 and 70°C, USB/Serial connection

Laptop camera – used as the main camera to fix hand movements in real time

Sensor Shield V5.0 – An expansion shield that makes it easy to connect sensors and motors to Arduino, providing a fast and convenient connection

Stepper motor – high-precision rotational motor, operating at 12V voltage

MG996R servo motor – powerful servo motor, with high torque, used for stronger and more precise control of fingers and mechanical parts

Technologies Used

Arduino (C/C++) – used to control robotic hand motors and mechanical parts.

Python is the main programming environment where all the analysis and AI work are done.

MediaPipe (Python library) detects 21 points in the hand and tracks their movement, understanding the position of the fingers.

OpenCV (Python library) is used to obtain images from a laptop camera and to process it in real time.

The KNN algorithm (Scikit-learn library) compares the positions of the hand points and understands the gesture displayed based on the patterns learned.

6. Results and Analysis

Project Results

As a result of our experiments, the robotic hand system has shown good and stable work in real-time mode.

The system achieves about 90% accuracy in gesture recognition, meaning it understands the position of the hand and the displayed gesture correctly for the most part

Tracking the 21 points of the hand using MediaPipe works stably and helps you correctly understand finger movements

The KNN model responds quickly enough, and the latency is small, meaning there is little delay between movement and robot response

There can be small errors in poor light conditions or with very fast movements

Analysis

The quality of the data from the camera has a big impact on the outcome

The better the data is cleaned (preprocessing), the more correctly the model works

Achieving the goal

- We have achieved the main goal of creating a robotic hand that is able to recognize and repeat the movements of the human hand in real time, as well as recognize gestures
 - The system works in real time
 - Sorting is done without direct human intervention
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7. Budget and Resources

Budget and Resources

□ *Devices and materials (expenses)*

Arduino Uno – Basic Control System

Notebook Camera — For Images

Sensor Shield V5.0 – for facilitating the connection of sensors and motors

MG996R servo motor – for robotic hand movements

Stepper motor – for basic mechanical rotations

Additional materials: metal and plastic constructions (for manual construction)

Connecting wires and power supplies

Software Resources (free)

Python (MediaPipe, OpenCV, Scikit-learn) – for AI and camera analysis

Arduino IDE – for microcontroller programming

Total costs

The project was carried out on a relatively low budget due to the use of learning and affordable components. It can be classified as a mid-value level DIY (do-it-yourself) project.

Human Resources

I have carried out this project entirely on my own. I did the programming, the electrical connections, the mechanical assembly and all the testing.

I worked on the project for about 2 months. During this time, there were intense phases and quieter working days.

I worked about 10 hours a day for about 3 weeks,

The rest of the time, about 1 to 2 hours a day for continuous improvement and experimentation.

Thus, the entire project has required a long and consistent work to develop the system at various stages and get a final working version.

8. Conclusion

✓ What Happened / What? You Reached

As a result of this work, I have developed a working prototype that is able to understand the position of the hand based on data from the camera and transmit it to the robotic hand.

The system works stably in real-time mode, recognizes various gestures and, in some cases, displays their names.

Overall, I have managed to combine the camera, AI models, and Arduino into a single system and get a practical result that shows how software and mechanical parts can be combined into a single working solution.

✓ What was innovative / what is special?

The innovation of this project is that different AI and visual technologies are combined in the system to achieve a single common result. A human simply shows the movement of the hand, and the system recognizes and reproduces it in real time with a robotic hand.

What is particularly unique is that the system does not use handheld physical control devices; the entire process is done solely on camera data.

In addition, the same system is also able to distinguish certain gestures and classify them, allowing it not only to replicate movements, but also to "understand" their type.

This results in a single, automated system that combines motion recognition, simulation, and simple AI-classification.

✓ What can be improved in the future?

In the future, the structure of the robotic arm could be improved by using higher quality and lighter materials to make it more robust and suitable for use.

In addition, the system can be made wearable, so that the hand is attached to a person's arm or forearm and immediately repeats the movements.

AI models and sensors could also be developed to make the system work more accurately in a variety of conditions and with fast movements.